Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Data Scientist and de facto team lead with 3 years of industry experience delivering analytics, ML and GenAI systems in regulated financial services, plus an MSc in Data Science (University of Bristol). Acted as de facto team lead for up to **4 interns** while staying hands-on, and worked directly with Client Service, Compliance and Investment from requirements to release. Delivered a **Python** pipeline that saves **60** analyst hours a month, **Apriori** market basket insights for marketing and **Power BI** dashboards that saved analysts **10** hours a week. Co-built the preprocessing pipeline and scoring algorithm for an MSc dissertation with AstraZeneca, validated with feedback from their scientists. **Python**, **R** and **SQL** on **AWS**.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Team Leadership & Stakeholders

- Built the first **GenAI** systems as the only engineer alongside my manager; as the team grew (a senior engineer and a frontend developer in year 2), acted as **de facto team lead** for **2 to 4 interns** in year 3, staying hands-on across design, code and deployment.
- Worked directly with **Client Service**, **Compliance**, **Investment** and **Marketing** from requirements to release; released to real users and iterated on their feedback, with prompts versioned in **Git** and **Compliance** reviewing every application; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).
- Made a **cost-aware model allocation** decision: **Llama-3** self-hosted on a GPU **EC2** instance for Client Service (the largest department) and **Azure OpenAI** for the other departments, keeping the better outputs for decision makers and the ability to switch models at scale.

Modelling, Analytics & Data

- Analyzed customer data with **Python Pandas** and **Market Basket Analysis** (**Apriori**), handing the marketing team insights on buying patterns; migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10 hours a week**, and recreated them in the **ReactJS** frontend so the wider organization could use the data without Power BI licences.
- Built a **Python** pipeline that parses over **300** daily emails from credit rating agencies into structured data, **fuzzy-matches** company names against those analysts track and sends a daily consolidated email, **saving 60 analyst hours a month**; first built on **AWS Textract**, then moved to **Azure OpenAI** when an agency began sending emails as images and broke the parser.
- Designed a **Normalized SQL Schema** (**PostgreSQL**) to the analysts' requirements and wrote **Python** ingestion scripts for the **Morningstar API**, covering every fund in the Indian mutual fund market (2023–2025), and integrated the legacy analyst data previously held in Excel into the same database.

GenAI, Cloud & Governance

- Designed and built **LLM applications** end to end (**ReactJS** frontend, **Python FastAPI** microservices, **RAG** with **LlamaIndex** and **Azure OpenAI**) with a separate **pgvector** database per department; **tool-calling** agents run **SQL** queries over the fund data to compare funds and generate PDF reports on request.
- Worked under security and compliance constraints: **RBAC** with a **Python** authentication function against the company **LDAP** server, **AWS Security Groups** limiting access to company private IPs and coordination with the security team until the applications passed their **VAPT**; designed the **AWS** infrastructure (**EC2**, **S3**, **ECR**, **Lambda**) and automated deployment of the frontend and backend through a one-click **Jenkins** and **GitLab CI/CD** pipeline.

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; personal focus on the data preprocessing, scoring algorithm and RAG assistant. Preprocessing pipeline joins 17 raw and augmented inputs (DepMap, GEO, HPA) into 14 clean tables with identity resolution, missing-value handling, integrity checks and an audit manifest; scoring ranks cancer cell lines by target inclusion and exclusion genes across six evidence layers, with calibration across lineages and confidence tiers; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Study comparing CNN/ViT/ResNet encoders with symbolic, SBERT embedding-retrieval and LSTM sequence decoders for image descriptions on three synthetic datasets, over five seeds, scored with SBERT cosine similarity, BLEU and ROUGE-L and compared with BLIP/SigLIP and commercial VLMs, noting where the evaluation protocols differed.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **Wingman:** AI dating-coach web app built end to end (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM) with 108 LLM personas across 12 archetypes and a vision model scoring uploaded photos. Rate limits shaped the design: 12 calls instead of 108, serialised calls, results streamed to the UI and written to disk incrementally; LLM output treated as untrusted (validated with Pydantic, repaired) and cached by input hash. GitHub: https://github.com/Arishtanemi4/wingman.git

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** AWS (EC2, S3, ECR, Lambda, EventBridge, SageMaker, CloudFormation templates), Docker, Jenkins, GitLab (CI/CD), Power BI, Azure OpenAI, LlamaIndex, Nginx, Linux, LDAP, Git, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, fuzzywuzzy), R (dplyr, tidyverse, ggplot2), SQL, JavaScript/TypeScript (NodeJS express, ReactJS), Shell
- **Databases:** PostgreSQL (schema design, SQL functions and procedures, pgvector), SQLite